## MCU

U1
ESP32-WROOM-32E-N4
+3V3
GND
ESP_EN
ESP_IO0
ESP_IO0
ESP_TX
ESP_RX
RTC_INT
I2C_SDA
I2C_SCL
DAC_OUT
AMP_SD
BTN_MODE
BTN_UP
BTN_DOWN
VBAT_SENSE
C8 10u
C9 100n
R9 10k
C10 1u
R10 10k
SW2 RESET
SW3 BOOT

## CHARGER

U9 LTC4412
Q3_GATE
VBAT
VSYS
GND
U2 MCP73831-2
VBUS
VBAT
PROG
CHG_STAT
Q3 AO3401A
C1 4.7u
C2 4.7u
R3 2k
R4 1k
C22 100n
C3 10u
R7 100k
R8 100k
C7 100n
J2 LiPo 3.7V
D1 CHG red
D2 B5819W

## USB UART

U4 CH340C
+3V3
CH340_V3
ESP_RX
ESP_TX
USB_DP
USB_DN
DTR
RTS
Q1 MMBT3904
Q2 MMBT3904
C11 100n
C21 100n
R14 0
R12 10k
R13 10k

## AUDIO

U8 PAM8302A
VSYS
AMP_INP
AMP_INN
AMP_SD
SPK_P
SPK_N
FM_L
FM_R
AUDIO_MIX
DAC_OUT
C16 1u
C17 2.2n
R19 10k
R20 10k
R21 10k
R22 10k
R23 10k
C18 1u
C19 10u
C20 100n
R11 100k
J4 Speaker 8R 1W

## RTC

U5 PCF8563T
RTC_VDD
I2C_SCL
I2C_SDA
RCLK_32K
RTC_INT
C12 100n
BT1 CR1220
R17 10k
R18 10k
D3 BAT54C
Y1 32.768k

## USB POWER

J1 USB-C
VBUS
CC1
CC2
USB_DN
USB_DP
SHIELD
U7 USBLC6-2SC6
R1 5.1k
R2 5.1k

## REGULATOR

U3 AP2112K-3.3
C4 1u
C5 10u
SW1 POWER
PWR_LED
D4 PWR green
R24 2k

## I2C

J3 OLED SSD1306 0.96in
R15 4.7k
R16 4.7k

## FM

U6 RDA5807M
C13 100n
C14 10u
C15 100p
AE1 FM wire antenna ~75cm

## UI

SW4 UP
SW5 DOWN
SW6 MODE

## MECHANICAL

H1 M2.5
H2 M2.5
H3 M2.5
H4 M2.5

PWR_FLAG GND
PWR_FLAG VBUS
PWR_FLAG VSYS
PWR_FLAG RTC_VDD

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|---|---|
| **aipcb** | |
| Sheet: / | |
| File: fm_radio.kicad_sch | |
| **Title: FM/BT Pocket Radio** | |
| Size: A2 | Date: |
| KiCad E.D.A. 10.0.1 | **Rev: A** |
| | Id: 1/1 |